DESIGNING AI BASED PLATFORM FOR SHARING INFORMATION INCASE OF ETHIOPIAN`S FARMER

1 INTRODUCTION

Agriculture in Ethiopia is considered one of the critical pillars of the country's economic stability, as more than 70% of the population depends on agricultural production for food and livelihood. As EIAR (2025) noted, the variety of climates in the nation enables different agricultural systems, including highland staples comprising teff, barley, and wheat, to livestock-supported livelihoods in the lowlands. In spite of such potential in natural resources, persistent challenges continue to burden the sector, particularly in the limited assimilation of modern information technologies into agricultural production systems. The ineffective sharing of knowledge and innovations on agriculture among regions has been a pressing issue, where most of the good practices remain confined to a locality due to major linguistic barriers and inadequate communication infrastructure. This has precipitated into falling productivity levels over recent years, forcing the Ethiopian government to seek technological interventions through initiatives such as digital agriculture and AI-assisted farming support.

Agriculture is central to the economy of Ethiopia, yet farmers continue to face many challenges that reduce output and sustainability. These challenges are both practical and organizational, caused by restricted access to timely information, weak networks for sharing knowledge, and slow integration of modern technologies into age-old farming practices.

Problems in Ethiopia's Agricultural Information Sharing

- Language Barriers
  - ✓ Lack of multilingual platforms limits farmers' knowledge to only those in their linguistic groups.
- Fragmented Knowledge exchange
  - ✓ There is no cross-regional learning and innovation as indigenous practices remain confined to a local setting.
- Weak Communication Infrastructure
  - ✓ Reliance on word-of-mouth, radio, and overstretched extension services limits timely and wide-scale information flow.
- Under-resourced Extension Services
  - ✓ Extension agents cannot support all farmers sufficiently, making the gap between traditional and modern practices even wider.
- Digital Divide
  - ✓ Low levels of digital literacy, poor connectivity, and limited availability of inclusive platforms discourage the adoption of technology.
- Declining Productivity
  - ✓ Outdated techniques persist because of poor dissemination of knowledge, hence yields remain below potential.

- ❖ Limit Cross-Regional Collaboration
  Refers to the absence or weakness of structured opportunities for farmers from different regions to share knowledge, practices, and innovations with one another.

General objective

It designs and prototype an inclusive, AI-powered information-sharing platform that capture, curate, and disseminates localized agricultural knowledge across Ethiopian regions to improve farmer decision-making, productivity, and resilience.

Specific objective

- ❖ Knowledge capture and localization
  - ✓ Develop mechanisms for gathering indigenous and expert practice related to planting, soil, irrigation, post-harvest, from regions such as Welega and Gojjam. Organize them into reusable language-accessible contents.
- ❖ AI-driven recommendation engine
  - ✓ Develop models that match practices and advisories to farmer context - region, crop, season, soil, risk profile - for timely, personalized guidance.
- ❖ Facilitating Farmer Conferences and Cross-Regional Collaboration
  - ✓ the organized process of bringing together farmers from different regions
- ❖ Real-time information integration:
  - ✓ Integrate weather forecasts, pest alerts, and market price feeds to provide actionable and region-specific updates that improve day-to-day decisions
- ❖ Inclusive multilingual access:
  - ✓ Enable interfaces and content in Amharic, Afan Oromo, Tigrinya, and local dialects; support voice-first and low-text interactions for low literacy users.
- ❖ Mobile-first, low-bandwidth design
  - ✓ Peer review, reputation scores, moderator workflows-including extension workers-are established to validate contributions to prevent misinformation.
- ❖ Data governance and ethics
  - ✓ Define clear consent, privacy, and data ownership policies; ensure equitable representation and avoidance of regional or linguistic bias.
- ❖ Sustainability and scale strategy
  - ✓ Develop a sustainability model: public–private partnerships, cooperatives, NGOs; also, provide a roadmap to scaling across regions and crops.